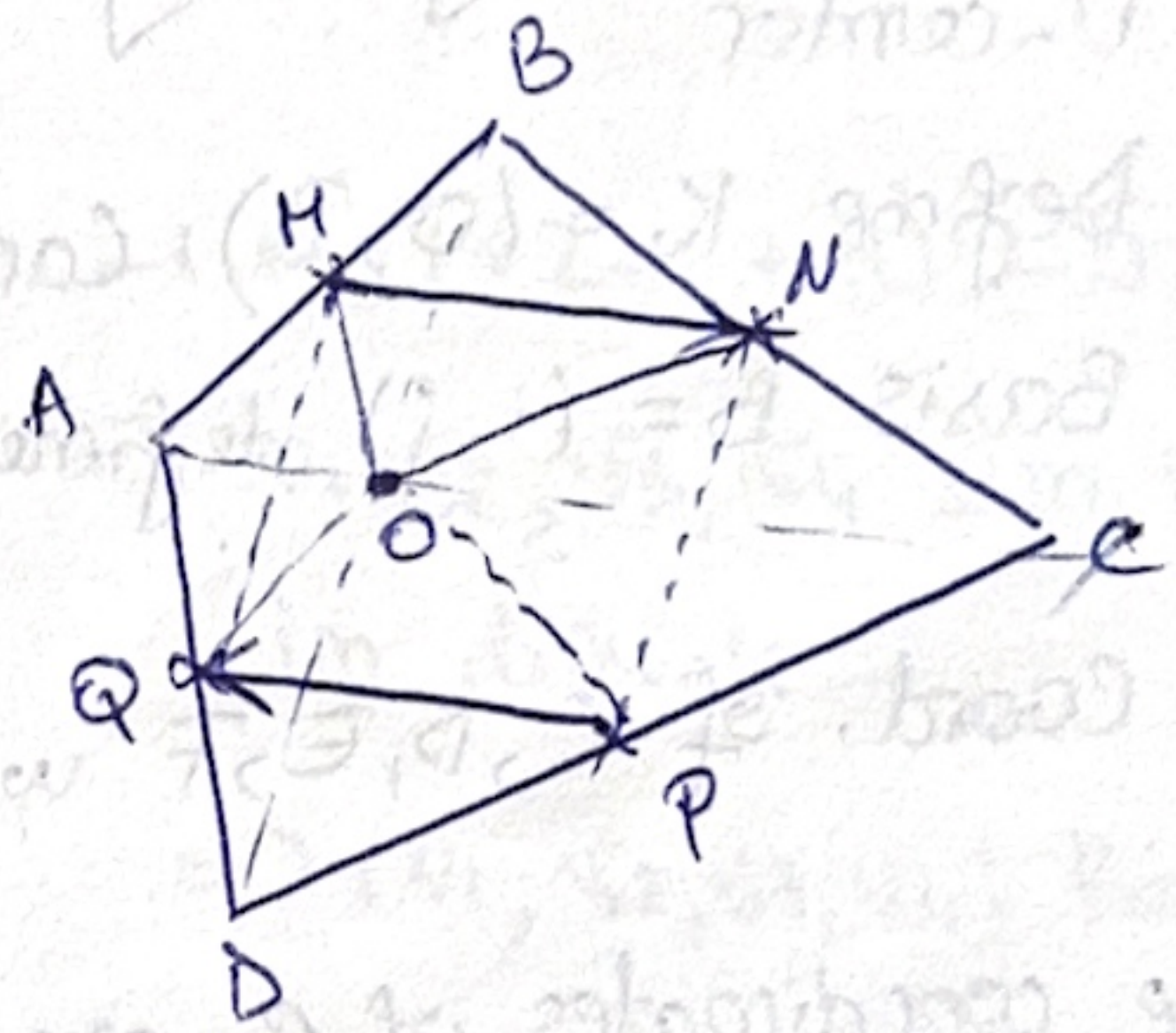


- ① ABCD quadrilateral
M, N, P, Q midpoints

$$\vec{MN} + \vec{PQ} = \vec{0}$$

MNPQ parallelogram?



$$\vec{MO} + \vec{ON} = \vec{MN}$$

$$\Leftrightarrow -\vec{OM} + \vec{ON} = \vec{MN}$$

$$\Leftrightarrow \vec{ON} - \vec{OM} = \vec{MN}$$

$$\vec{PO} + \vec{OQ} = \vec{PQ}$$

$$\Leftrightarrow \vec{OQ} - \vec{OP} = \vec{PQ}$$

$$\begin{aligned} \vec{MN} + \vec{PQ} &= \vec{ON} - \vec{OM} + \vec{OQ} - \vec{OP} = \frac{\vec{OB} + \vec{OC}}{2} - \frac{\vec{OB} + \vec{OA}}{2} + \frac{\vec{OA} + \vec{OD}}{2} - \frac{\vec{OC} + \vec{OD}}{2} \\ &= \frac{\vec{OB} + \vec{OC} - \vec{OB} - \vec{OA} + \vec{OA} + \vec{OD} - \vec{OC} - \vec{OD}}{2} = \vec{0} \end{aligned}$$

$$\Rightarrow \vec{MN} + \vec{PQ} = \vec{0} \Rightarrow \vec{MN} = -\vec{PQ} \Rightarrow \vec{MN} = \vec{QP} \Rightarrow \text{MNPQ parallelogram.}$$

- ② ABCD quadr.

E - mid. [AB]

F - mid. [CD]

$$\vec{EF} = \frac{1}{2} (\vec{AD} + \vec{BC})$$

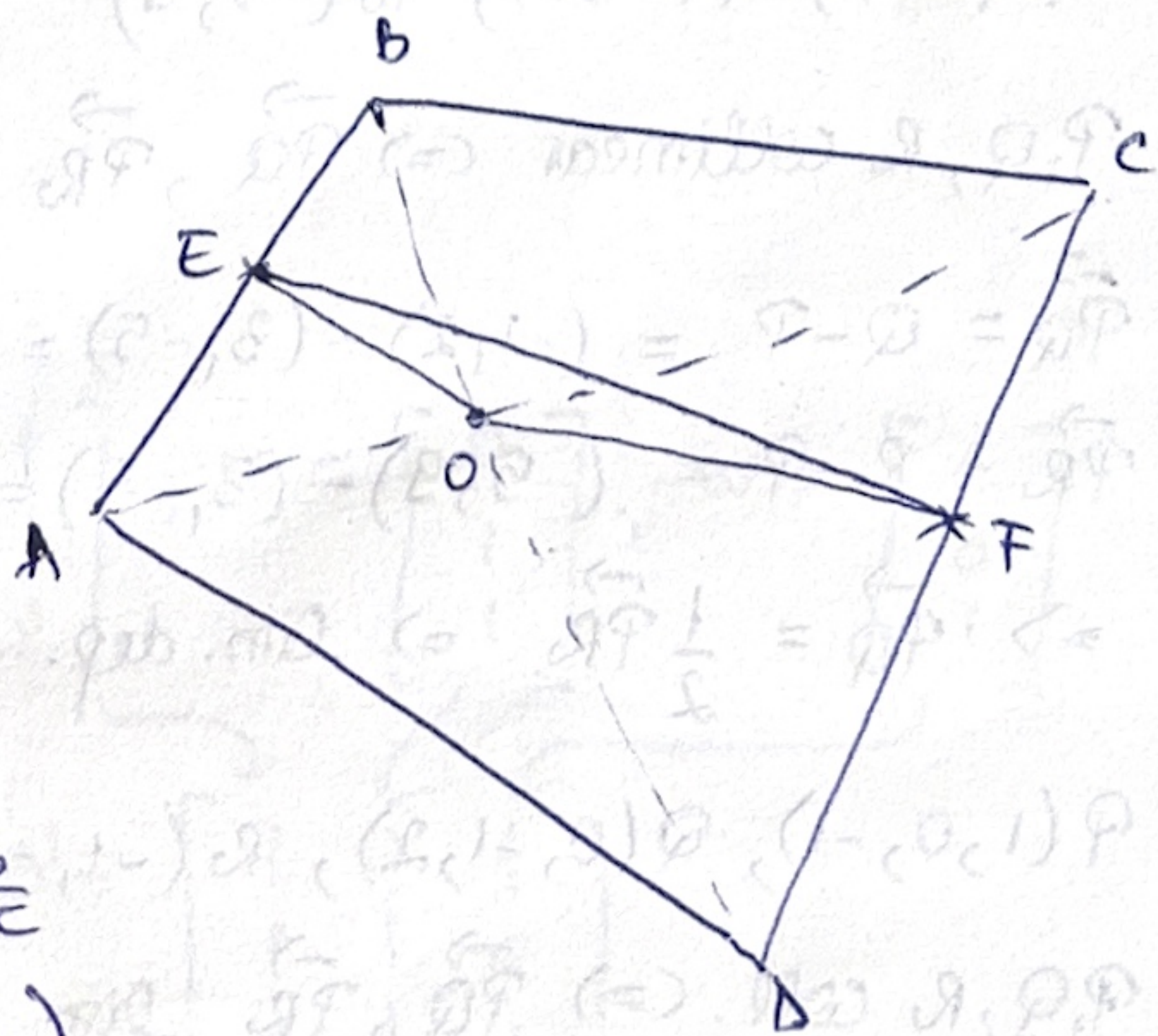
$$\vec{EO} + \vec{OF} = \vec{EF}$$

$$\Leftrightarrow -\vec{OE} + \vec{OF} = \vec{EF} \Leftrightarrow \vec{EF} = \vec{OF} - \vec{OE}$$

$$\Leftrightarrow \vec{EF} = \frac{1}{2} (\vec{OC} + \vec{OD}) - \frac{1}{2} (\vec{OA} + \vec{OB})$$

$$= \frac{1}{2} (\vec{OD} + \vec{OC} - \vec{OA} - \vec{OB})$$

$$= \frac{1}{2} (\vec{AD} + \vec{BC})$$



$$\vec{AD} + \vec{OD} = \vec{AD}$$

$$\Rightarrow \vec{AD} = \vec{OD} - \vec{OA}$$

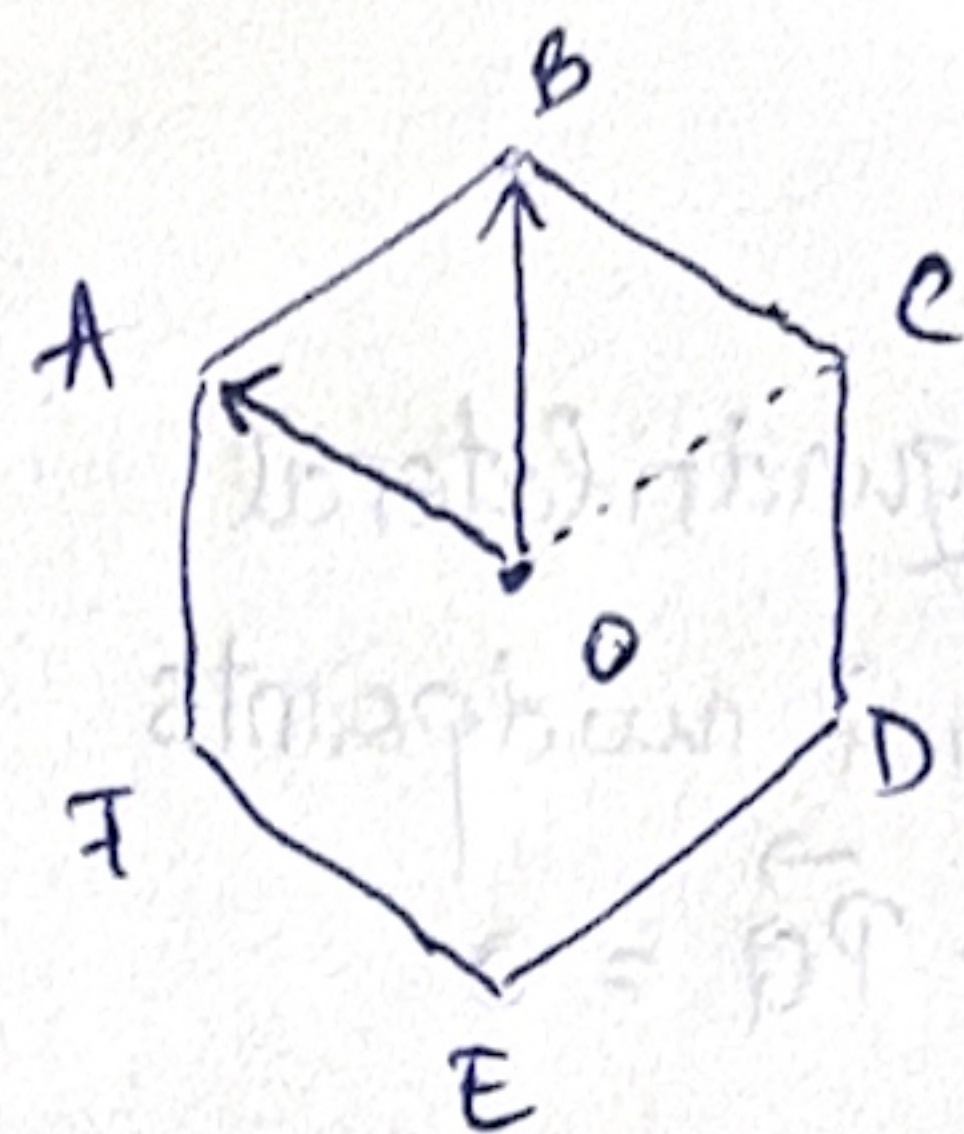
$$\vec{BC} + \vec{OC} = \vec{BC}$$

$$\Rightarrow \vec{BC} = \vec{OC} - \vec{OB}$$

! If ABCD trapezoid, then $\vec{EF}$, $\vec{AD}$, $\vec{BC}$ are collinear and define the same direction

$$|\vec{EF}| = \left| \frac{1}{2} (\vec{AD} + \vec{BC}) \right| = \frac{1}{2} (|\vec{AD}| + |\vec{BC}|)$$

- ③ plane $\mathbb{E}^2$
 ABCDEF hexagon reg.
 O - center



- Define $K = (O, B)$ Cartesian frame
 Basis $B = (i, j)$ defined by $i = \vec{OA}$
 $j = \vec{OB}$
- Coord. of C, D, E, F w.r. to K.

The coordinates of C are the components of its position vector $\vec{OC}$.

$$\vec{OC} = \vec{OB} + \vec{BC} \xrightarrow{\text{OABC par.}} \vec{OB} + \vec{AO} = -\vec{OA} + \vec{OB} = -i + j$$

$\Rightarrow C_K (-1, 1) \rightarrow$ cartesian frame of C

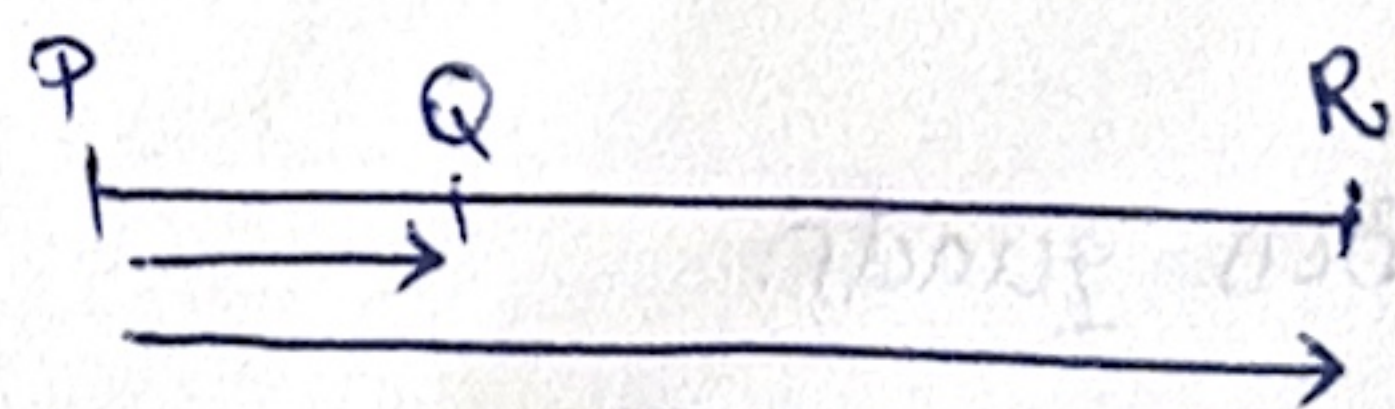
$$\vec{OD} = -\vec{OA} = -i \Rightarrow D_K (-1, 0) \rightarrow \text{cartesian frame of D}$$

$$\vec{OE} = -\vec{OB} = -j \Rightarrow E_K (0, -1) \rightarrow \text{cartesian frame of E}$$

$$\vec{OF} = -\vec{OC} = i - j \Rightarrow F_K (1, -1) \rightarrow \text{cartesian frame of F}$$

- ④ Collinear?

a) $P(3, -5), Q(-1, 2), R(-5, 9)$



P, Q, R collinear $\Leftrightarrow \vec{PQ}, \vec{PR}$ lin. dep.

$$\vec{PQ} = Q - P = (-1, 2) - (3, -5) = (-4, 7) \Rightarrow \vec{PQ} (-4, 7)$$

$$\vec{PR} = R - P = (-5, 9) - (3, -5) = (-8, 14) \Rightarrow \vec{PR} (-8, 14)$$

$$\Rightarrow \vec{PQ} = \frac{1}{2} \vec{PR} \Rightarrow \text{lin. dep.} \Rightarrow \text{collinear} \checkmark$$

b) $P(1, 0, -1), Q(0, -1, 2), R(-1, -2, 5)$

P, Q, R coll. $\Leftrightarrow \vec{PQ}, \vec{PR}$ lin. dep.

$$\vec{PQ} (-1, -1, 3) \quad \vec{PR} (-2, -2, 6) \quad \Rightarrow \vec{PQ} = \frac{1}{2} \vec{PR} \Rightarrow \vec{PR} = 2\vec{PQ} \Rightarrow \text{lin. dep.} \Rightarrow \text{coll.}$$

- ⑤ Form a basis?

a) $v_1(1, 2), v_2(3, 4)$

$$v_1, v_2 \text{ not proportional } \left(\frac{1}{3} \neq \frac{2}{4} = \frac{1}{2} \right) \Rightarrow \text{lin. ind.}$$

We see two components $\Rightarrow$ ambient vector space has dim 2

$\Rightarrow (v_1, v_2)$ basis.

0) $N_1(-1, 2, 1, 0)$, $N_2(2, 1, 1, 0)$, $N_3(1, 0, -1, 1)$, $N_4(1, 0, 0, 1)$

check $\det \neq 0$?

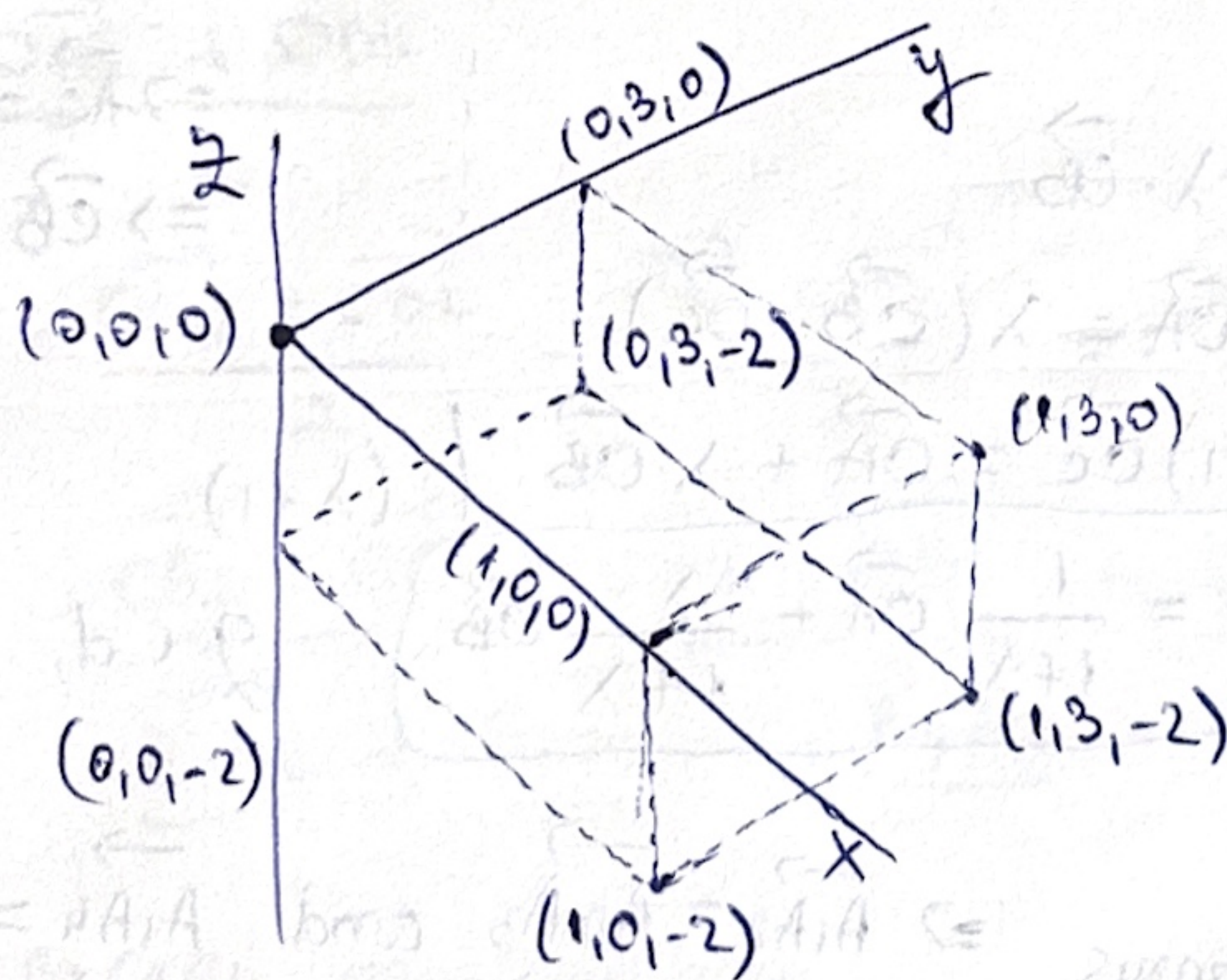
?

$$\begin{vmatrix} -1 & 2 & 1 & 1 \\ 2 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{vmatrix} = - \begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & -1 \\ 0 & 0 & 1 \end{vmatrix} + \begin{vmatrix} -1 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & -1 \end{vmatrix} = -1 + (1+2-1+4) = 5 \neq 0$$

$\Rightarrow N_1, N_2, N_3, N_4$ lin. ind.
in dim. 4

$\Rightarrow (N_1, N_2, N_3, N_4)$ Basis $\checkmark$.

⑥ Coord. of the vertices of the parallelepiped whose faces lie in the coord. planes and in the planes $x=1$, $y=3$, $z=-2$.



How to draw?

⑦ $B = (i, j, k)$ basis.

$$u = i + j$$

$$v = j + k$$

$$w = i + k$$

$B' = (u, v, w)$? basis

$M_{B'B}, M_{BB'} = ?$

$$u = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, v = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, w = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$M_{BB'} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$M_{BB'}$

$$\det M_{BB'} = 2 \neq 0 \Rightarrow \text{basis} \Rightarrow$$

$\Rightarrow B, B'$ have the same orientation

(if B is left oriented then so is B')

$M_{B'B} = M_{BB'}^{-1} \rightarrow$ calculate OR express i, j, k in terms of u, v, w .

$$u = i + j \Rightarrow i = u - j \Rightarrow \begin{cases} v = j + k \\ w = u - j + k \end{cases}$$

$$+ \quad v + w = u + 2k$$

$$- \quad v - w = -u + 2j \Rightarrow j = \frac{1}{2}(u + v - w)$$

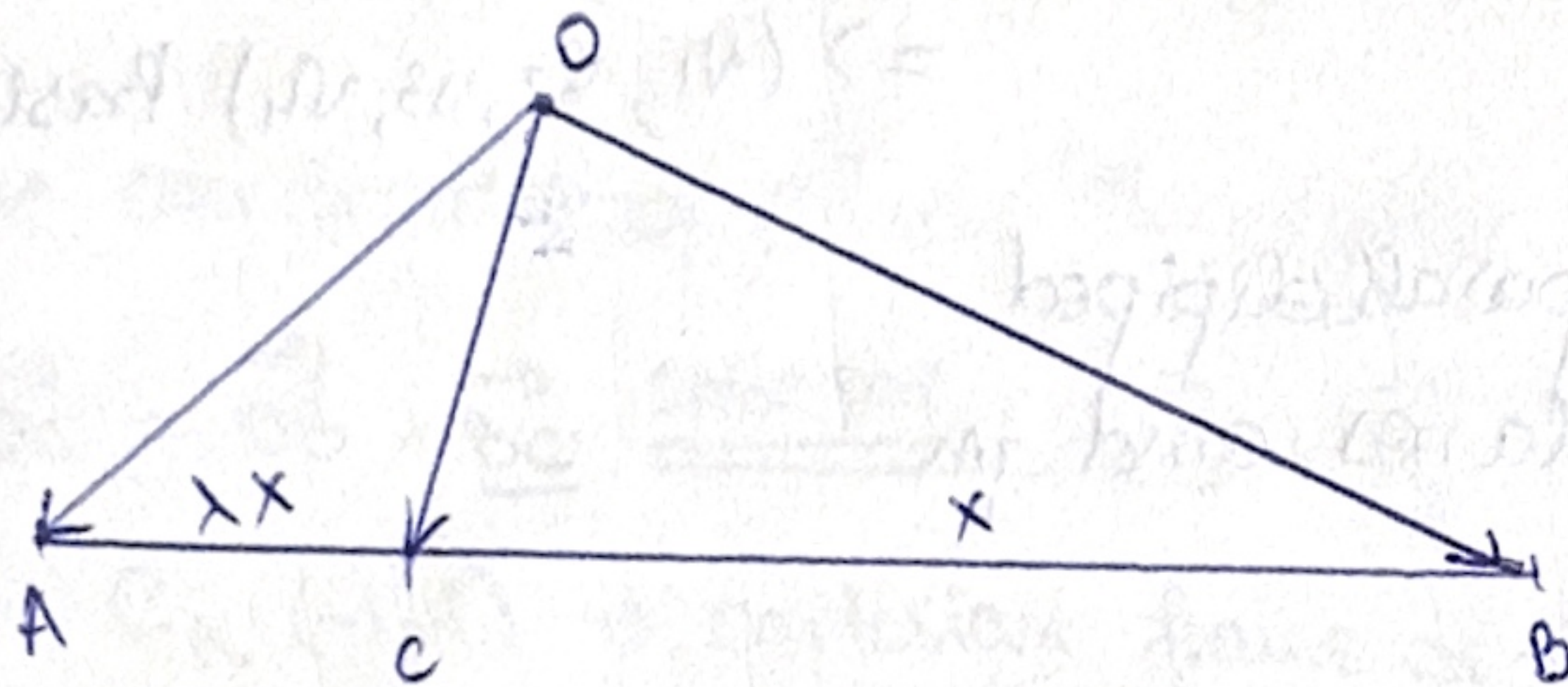
$$\Rightarrow i = \frac{1}{2}(u - v + w)$$

$$\Rightarrow M_{B'B} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$

7, 10 seminars present!

(1.1) H: $\lambda = |CA| : |CB|$, $C \in [AB]$

cl: $\vec{OC} = \frac{1}{1+\lambda} \vec{OA} + \frac{\lambda}{1+\lambda} \vec{OB}$



A-C-B
"collinear"

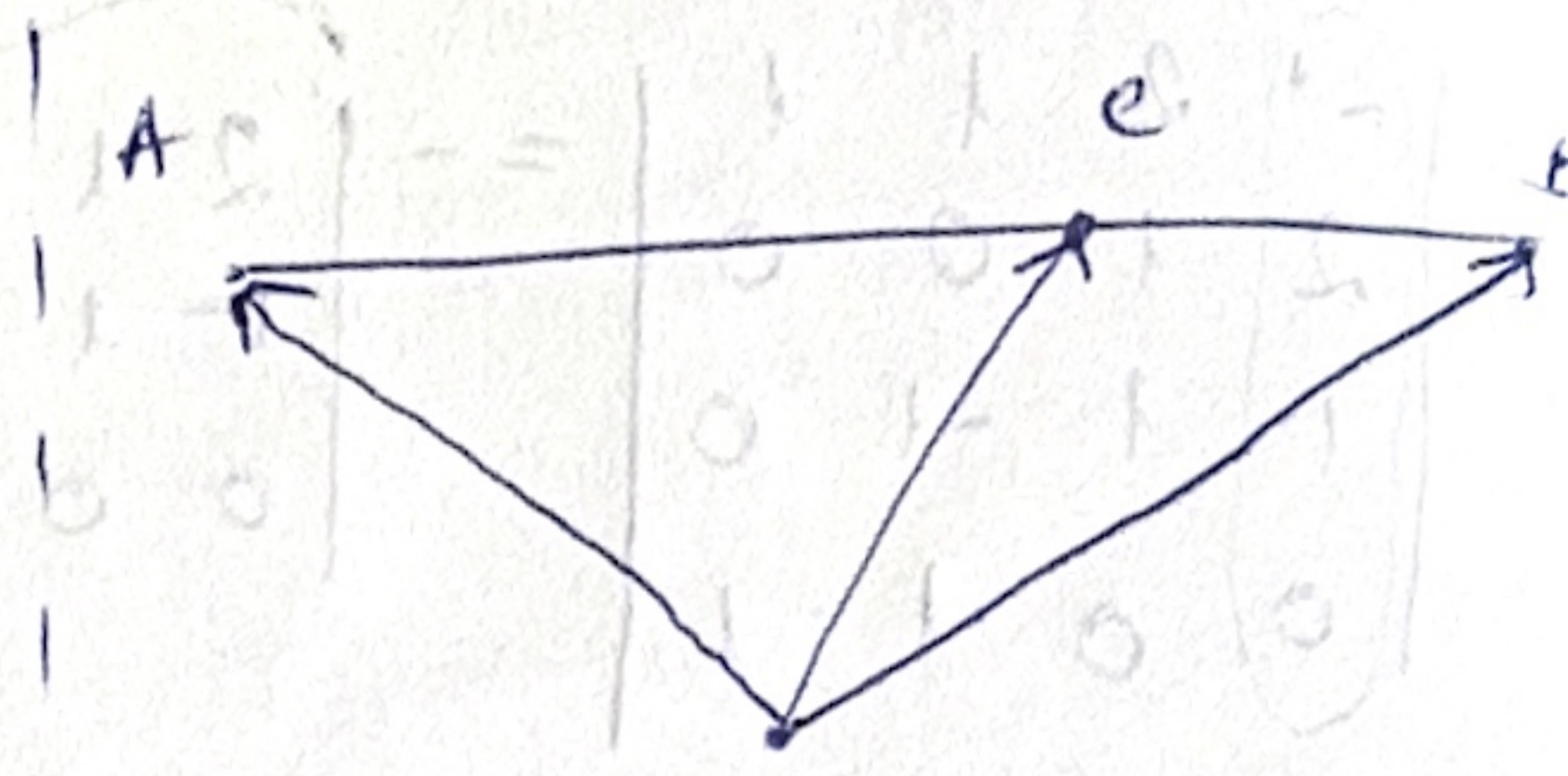
$\lambda = |CA| : |CB|$

$\Rightarrow \vec{AC} = \lambda \cdot \vec{CB}$

$(\Rightarrow) \vec{OC} - \vec{OA} = \lambda (\vec{OB} - \vec{OC})$

$(\Rightarrow) (\lambda + 1) \vec{OC} = \vec{OA} + \lambda \cdot \vec{OB} \quad | : (\lambda + 1)$

$(\Rightarrow) \vec{OC} = \frac{1}{1+\lambda} \vec{OA} + \frac{\lambda}{1+\lambda} \vec{OB} \quad \text{g.e.d.}$



$\frac{CA}{CB} = \lambda$

$\Rightarrow CA = \lambda \cdot CB$

$\vec{OC} = \vec{OA} + \vec{AC}$

$\vec{OC} = \vec{OB} + \vec{BC}$

$\Rightarrow \vec{AC} = \vec{OC} - \vec{OA}$

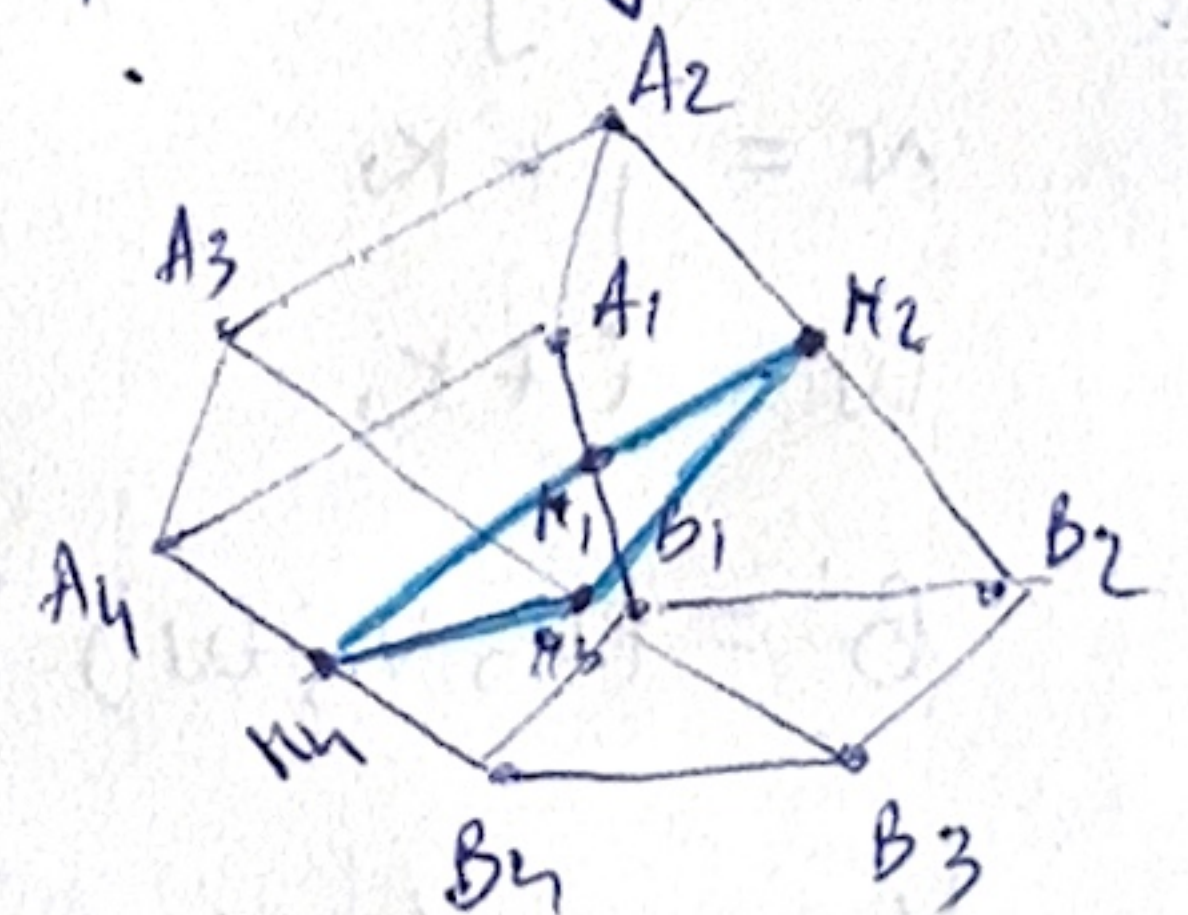
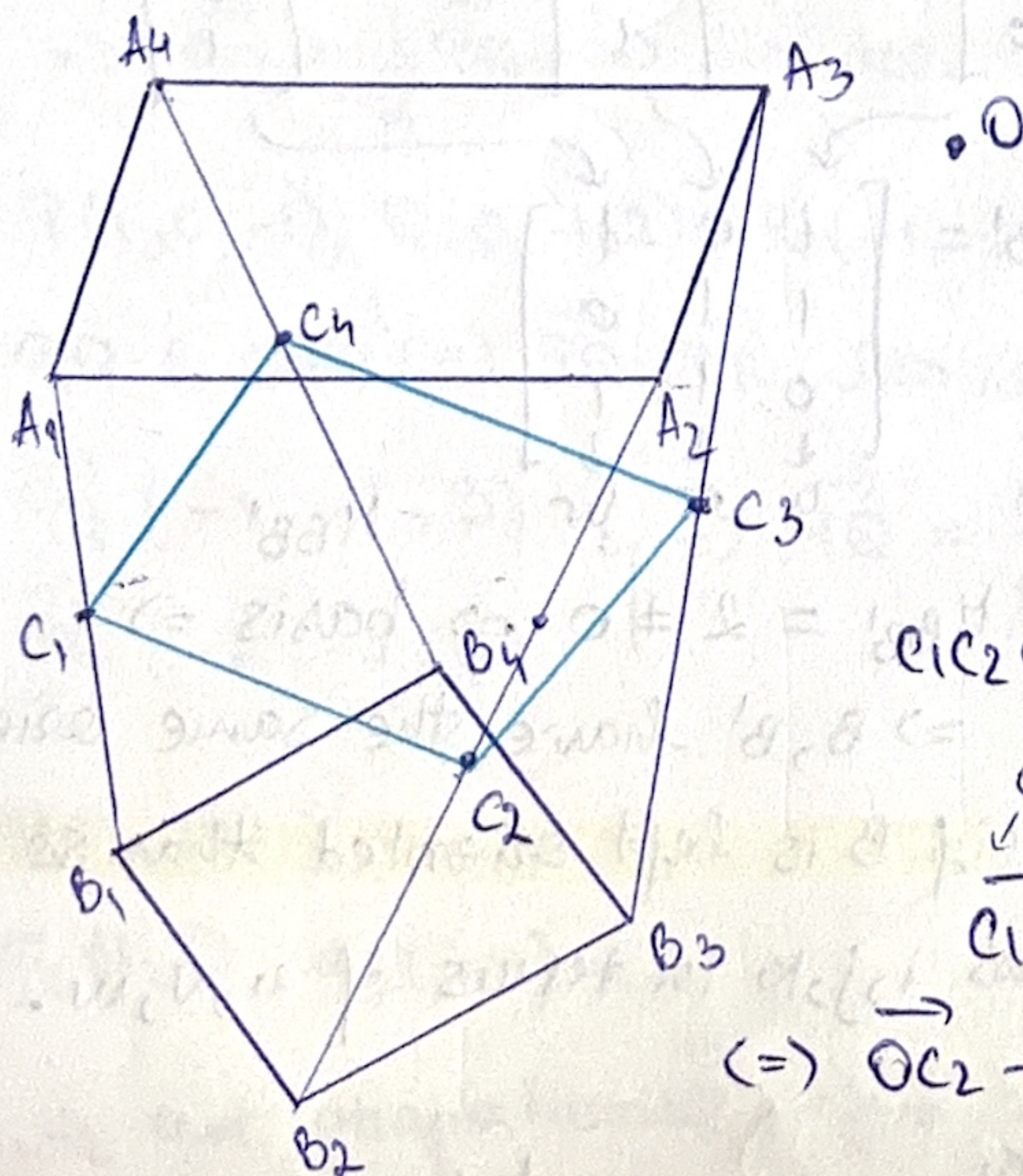
$\Rightarrow \vec{CB} = \vec{OB} - \vec{OC}$

(1.2) A_1, A_2, A_3, A_4 } parallelograms.
 B_1, B_2, B_3, B_4 }

$\Rightarrow \vec{A_1A_2} = \vec{A_4A_3}$ and $\vec{A_1A_4} = \vec{A_2A_3}$ (1)

$\Rightarrow \vec{B_1B_2} = \vec{B_4B_3}$ and $\vec{B_1B_4} = \vec{B_2B_3}$

cl: midpoints of $[A_1B_1], [A_2B_2], [A_3B_3], [A_4B_4] \Rightarrow$ parallelogram.

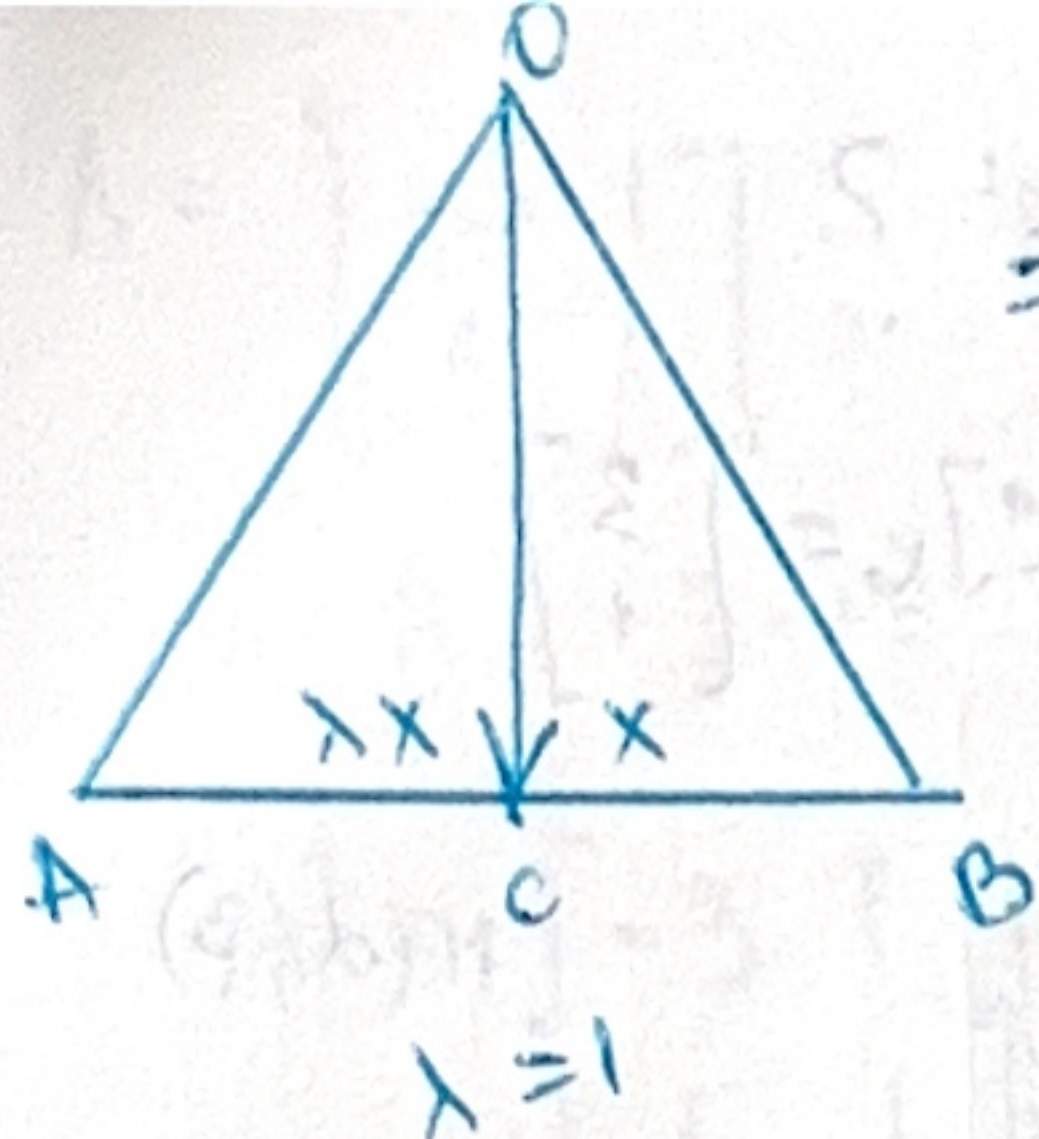


$C_1C_2C_3C_4$ parallelogram $(\Rightarrow) \vec{C_1C_2} = \vec{C_4C_3}$

$(\Leftarrow) \vec{C_4C_1} = \vec{C_3C_2}$

$\vec{C_1C_2} = \vec{C_4C_3} (*)$

$(\Rightarrow) \vec{OC_2} - \vec{OC_1} = \vec{OC_3} - \vec{OC_4}$, O - arbitrary fixed point.



$$\Rightarrow \vec{OC} = \frac{\vec{OA} + \vec{OB}}{2} \quad \text{"Median theorem in triangle"}$$

$$\vec{OC}_2 = \frac{\vec{OA}_2 + \vec{OB}_2}{2}$$

$$\vec{OC}_1 = \frac{\vec{OA}_1 + \vec{OB}_1}{2}$$

$$\vec{OC}_3 = \frac{\vec{OA}_3 + \vec{OB}_3}{2}$$

$$\vec{OC}_4 = \frac{\vec{OA}_4 + \vec{OB}_4}{2}$$

$$(*) (2) \quad \frac{\vec{OA}_2 + \vec{OB}_2}{2} - \frac{\vec{OA}_1 + \vec{OB}_1}{2} = \frac{\vec{OA}_3 + \vec{OB}_3}{2} - \frac{\vec{OA}_4 + \vec{OB}_4}{2} \quad / \cdot 2$$

$$(2) \quad \vec{OA}_2 - \vec{OA}_1 + \vec{OB}_2 - \vec{OB}_1 = \vec{OA}_3 - \vec{OA}_4 + \vec{OB}_3 - \vec{OB}_4$$

$$(2) \quad \vec{A_1A_2} + \vec{B_1B_2} = \vec{A_4A_3} + \vec{B_4B_3} \quad \text{True because (1).}$$

$\Rightarrow C_1C_2C_3C_4$ parallelogram.

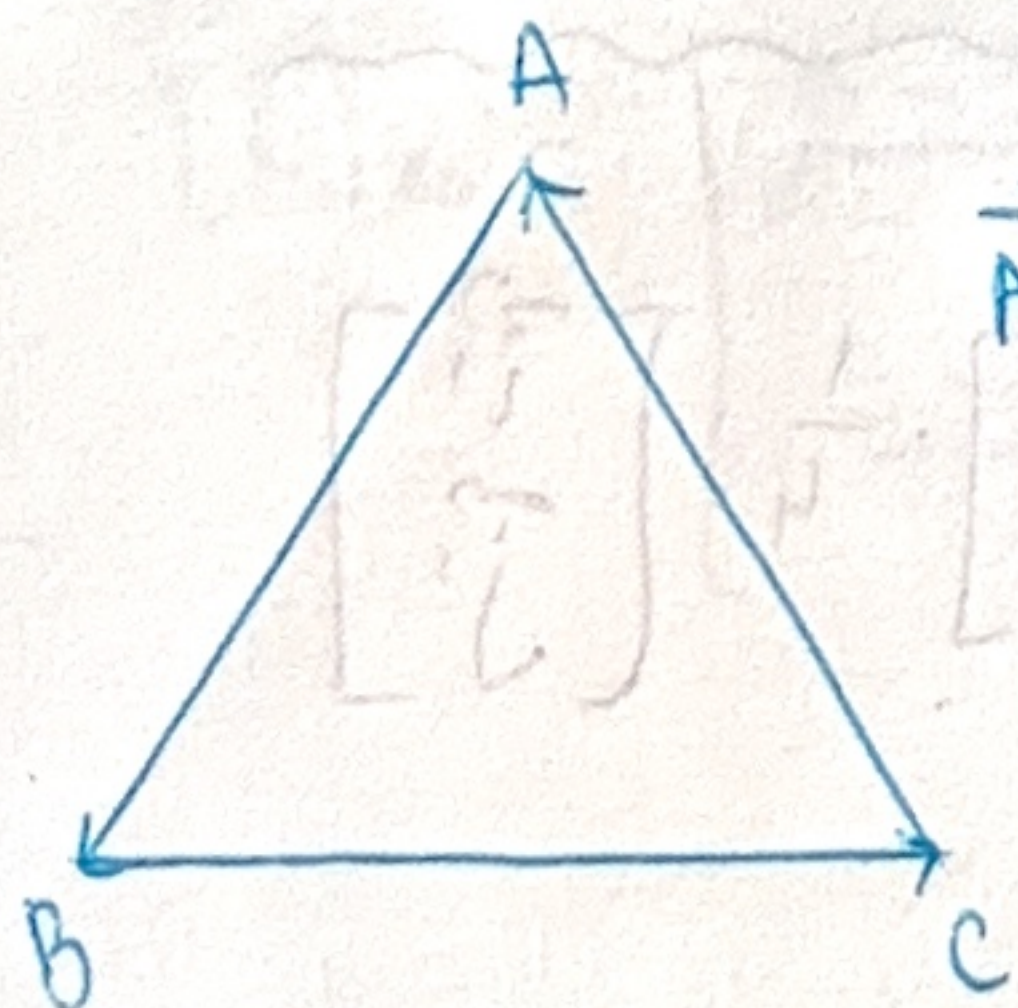
! A parallelogram can be characterised by equal vector equations (2).

1.3. u, v, w be vertices of Δ ?

a) $u(7,3), v(2,8), w(-5,5)$

$$uv + vw > uw$$

b) $u(1,2,-1), v(2,-1,0), w(1,-3,1)$



$$\vec{AB} + \vec{BC} + \vec{CA} = \vec{0}$$

$$uv(-5,5)$$

$$vw(-4,-3)$$

$$uw(-4,-3)$$

$$uv(-4,-3)$$

$$uw(-4,-3)$$

a) $\vec{u} + \vec{w} = \vec{v} \Rightarrow \vec{u} + (-\vec{v}) + \vec{w} = \vec{0} \Rightarrow$ they form a triangle.

b) $\vec{u} + \vec{w} = \vec{v} \Rightarrow$ " " " "

OR with determinant:

$$\Rightarrow 2 \text{ equal lines} \Rightarrow \det = 0.$$

(1.4)

$$K = (0, \beta)$$

$$K' = (0', \beta')$$

• Base change matrix from $K \rightarrow K'$?

$$[A]_K = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, [B]_K = \begin{bmatrix} 6 \\ 4 \end{bmatrix}, [C]_K = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$\beta = (i, j)$$

$$\beta' = (i', j')$$

$$[0']_K = \begin{bmatrix} 6 \\ 0 \end{bmatrix} \Rightarrow 00' = 6\vec{i} + 0\vec{j} \quad (2)$$

$$[i']_\beta = \begin{bmatrix} -3 \\ 1 \end{bmatrix} \Rightarrow i' = -3\vec{i} + 1\vec{j}$$

$$[j']_\beta = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow j' = 1\vec{i} + 1\vec{j}$$

$$\Rightarrow \begin{bmatrix} \vec{i}' \\ \vec{j}' \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \vec{i} \\ \vec{j} \end{bmatrix}$$

$$[H]_K = \alpha\vec{i} + \beta\vec{j} \quad \vec{OH} = \alpha\vec{i} + \beta\vec{j} \quad (1)$$

$$[H]_K = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$[H]_{K'} = ?$$

$$\vec{O'H} = \vec{O'O} + \vec{OH} = -\vec{OO'} + \vec{OH}$$

$$\stackrel{(1)}{=} -(6\vec{i} + 0\vec{j}) + \alpha\vec{i} + \beta\vec{j}$$

$$\stackrel{(2)}{=} (\alpha - 6)\vec{i} + \beta\vec{j}$$

$$= [\alpha - 6 \quad \beta] \cdot \begin{bmatrix} \vec{i}' \\ \vec{j}' \end{bmatrix} \quad (*)$$

$$A^{-1} \cdot \underbrace{\begin{bmatrix} \vec{i}' \\ \vec{j}' \end{bmatrix}}_A = \begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} \vec{i} \\ \vec{j} \end{bmatrix}}_A \Rightarrow \underbrace{\begin{bmatrix} \vec{i}' \\ \vec{j}' \end{bmatrix}}_A = A^{-1} \cdot \underbrace{\begin{bmatrix} \vec{i} \\ \vec{j} \end{bmatrix}}_A$$

$$(*) : \vec{O'H} = [\alpha - 6 \quad \beta] \cdot A^{-1} \cdot \begin{bmatrix} \vec{i}' \\ \vec{j}' \end{bmatrix} = [\alpha - 6 \quad \beta] \cdot \begin{bmatrix} -1 & 1 \\ 1 & 3 \end{bmatrix} \cdot \frac{1}{4} \cdot \begin{bmatrix} \vec{i}' \\ \vec{j}' \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 6 - \alpha + \beta & \alpha - 6 + 3\beta \end{bmatrix} \cdot \begin{bmatrix} \vec{i}' \\ \vec{j}' \end{bmatrix}$$

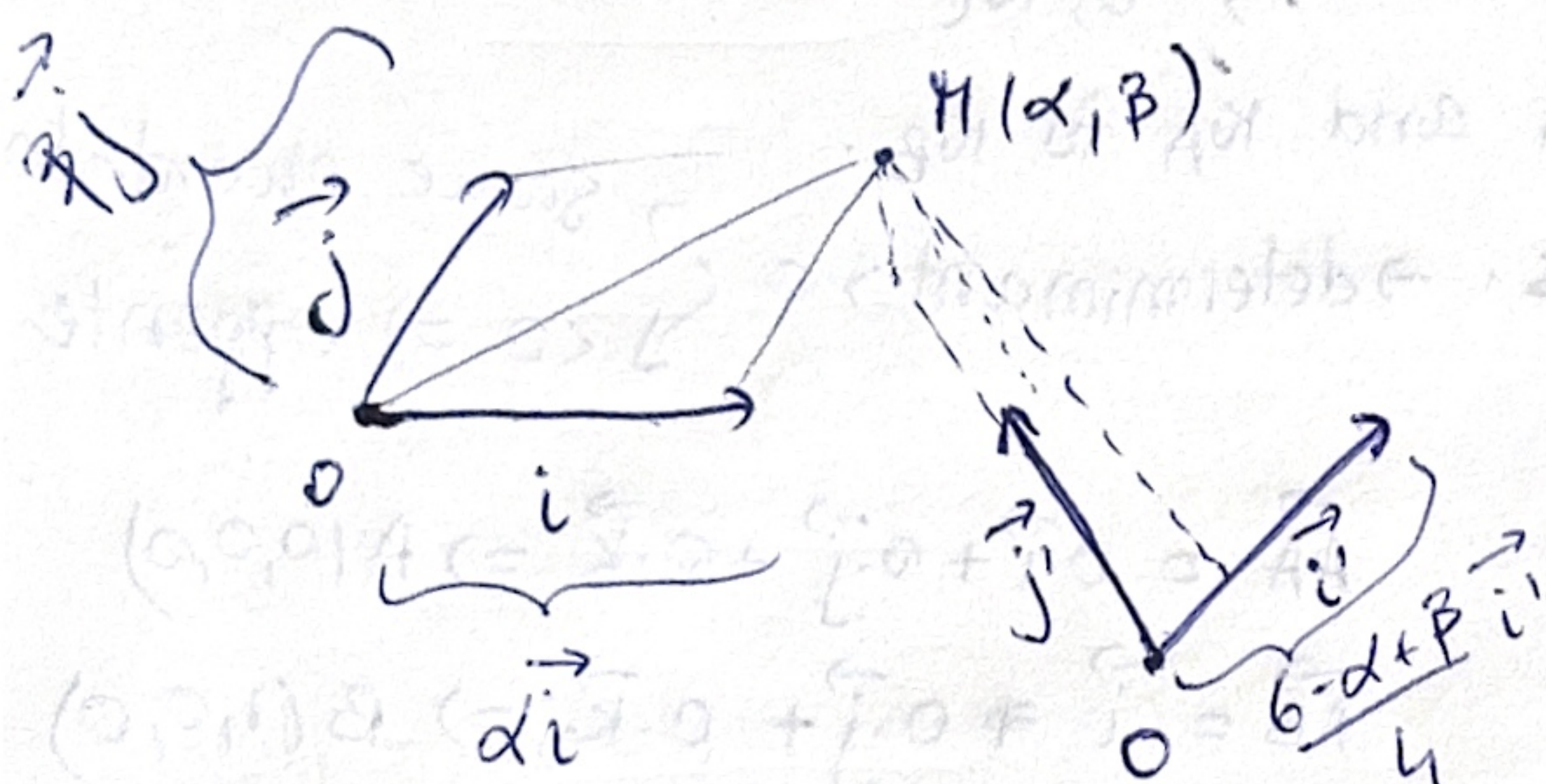
$$= \frac{1}{4} \left((6 - \alpha + \beta)\vec{i}' + (\alpha - 6 + 3\beta)\vec{j}' \right)$$

↳ formula for changing from one frame to another.

$$A = \begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix}; A^{-1} = ? \quad A^{-1} = \frac{1}{\det A} \cdot A^* = \frac{1}{-4} \cdot \begin{bmatrix} 1 & -1 \\ -1 & -3 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -1 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\det A = -3 - 1 = -4$$

$$A^t = \begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow a_{11}^* = 1 \quad a_{21}^* = 1 \\ a_{12}^* = 1 \quad a_{22}^* = -3$$



$[H_{KK'}] = A^{-1}$.
 The matrix of change of basis from K to K' ? $\rightarrow$ always true or only here? always true.

$$[H]_K = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \Rightarrow [H]_{K'} = \begin{bmatrix} \frac{6-\alpha+\beta}{4} \\ \frac{\alpha-6+3\beta}{4} \end{bmatrix} \rightarrow \text{use these for } A, B, C.$$

$$[A]_{K'} = \begin{bmatrix} \frac{6-0+2}{4} \\ \frac{0-6+3 \cdot 2}{4} \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$[B]_{K'} = \begin{bmatrix} \frac{6-6+4}{4} \\ \frac{6-6+3 \cdot 4}{4} \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$[C]_{K'} = \begin{bmatrix} \frac{6-3+1}{4} \\ \frac{3-6+3}{4} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

1.5) ABCD tetrahedron

$$K_A = (A, \beta), \quad \beta = (\vec{AB}, \vec{AC}, \vec{AD})$$

$$K_B = (B, \beta), \quad \beta = (\vec{BA}, \vec{BC}, \vec{BD})$$

$$K_A' = (A, \beta), \quad \beta = (\vec{BA}, \vec{BD}, \vec{BC})$$

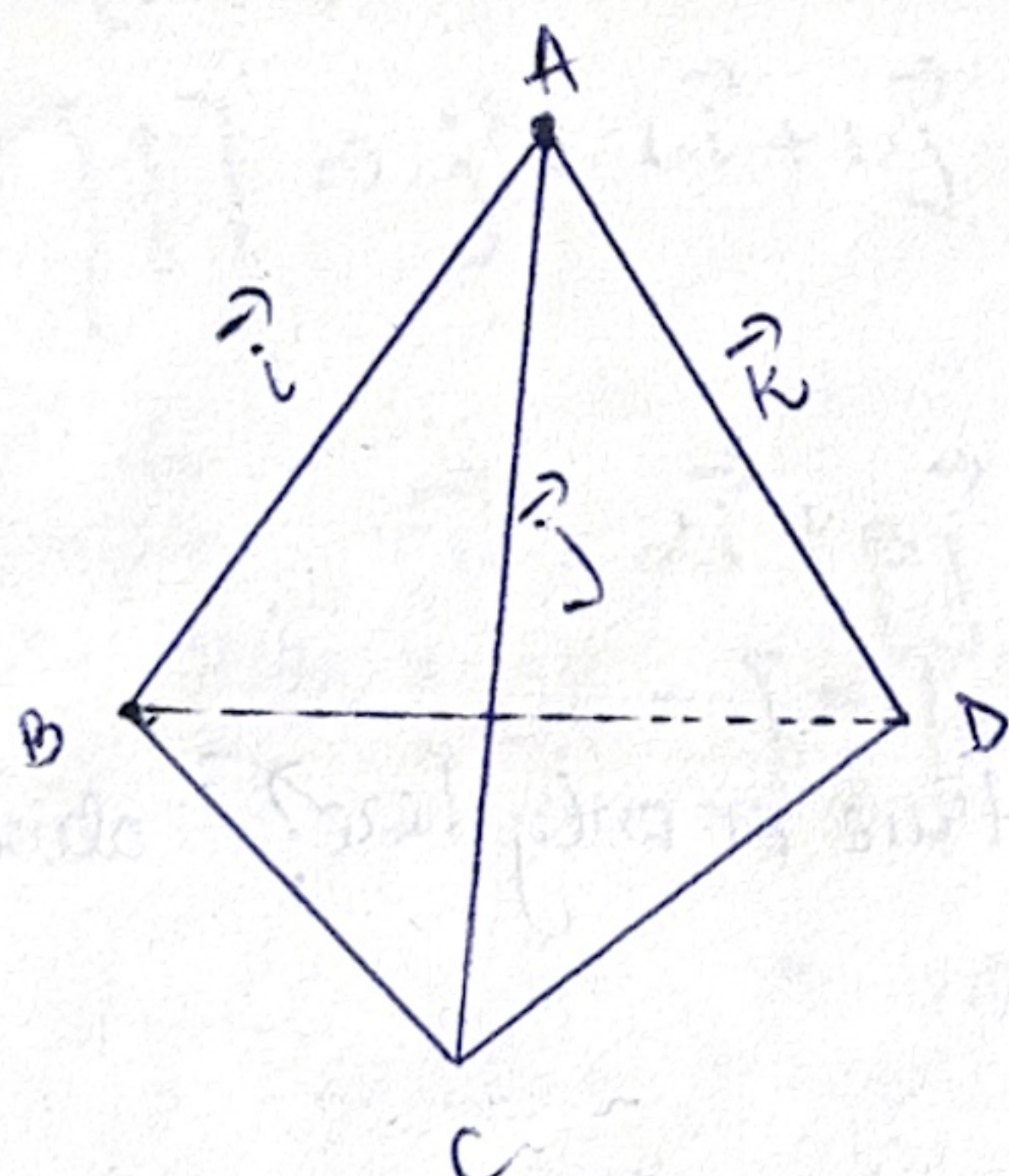
a) $A, B, C, D = (\dots, \dots, \dots)$ w.r. to K_A, K_B, K_C

b) base change matrix K_A' to K_A and K_A to K_B .

c) relative orientation of frames. $\rightarrow$ determinant > 0 $\rightarrow$ same orientation
 $< 0 \Rightarrow$ opposite

a)

K_A :



$$\vec{AA} = 0 \cdot \vec{i} + 0 \cdot \vec{j} + 0 \cdot \vec{k} \Rightarrow A(0, 0, 0)$$

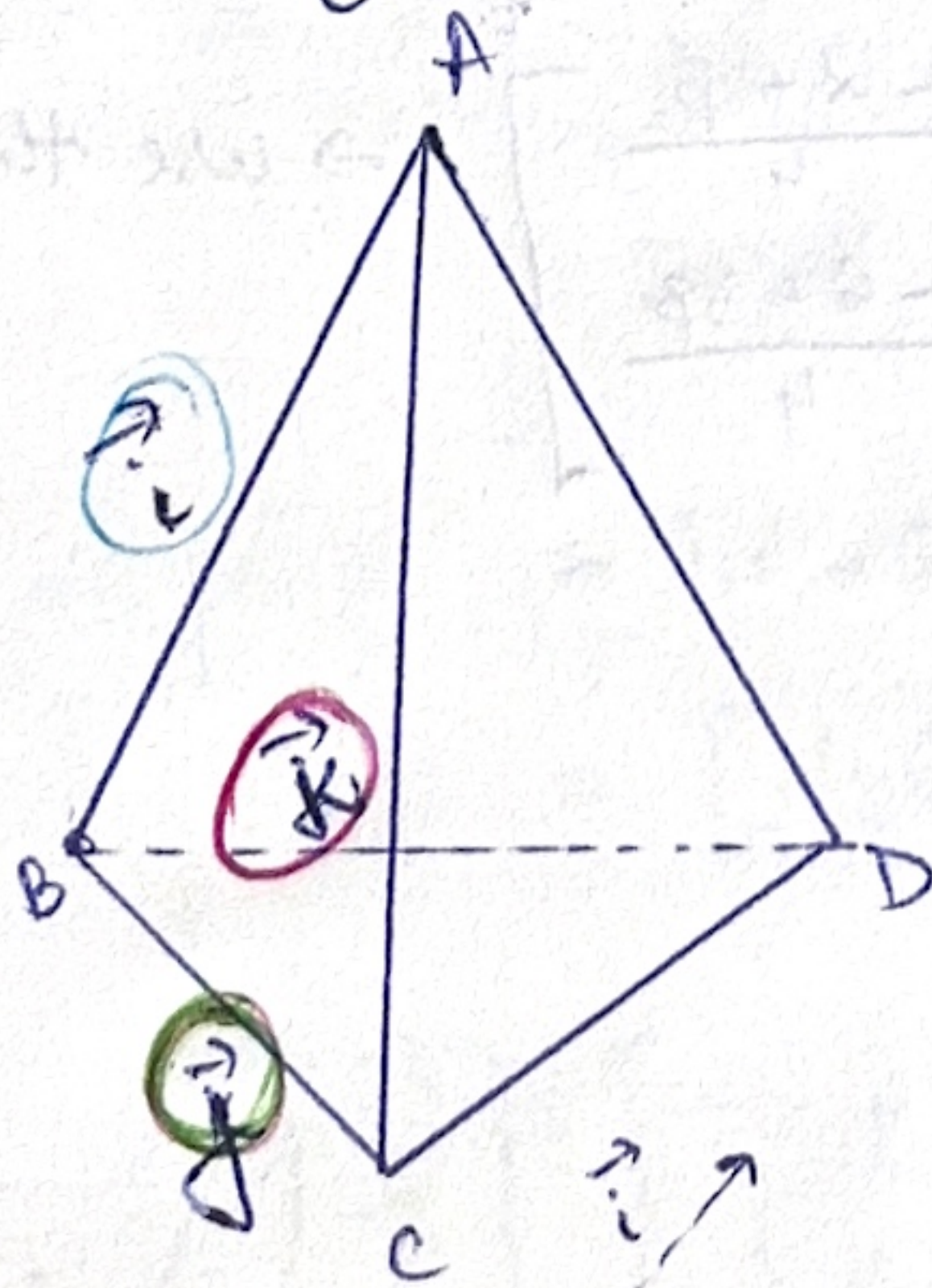
$$\vec{AB} = 1 \cdot \vec{i} + 0 \cdot \vec{j} + 0 \cdot \vec{k} \Rightarrow B(1, 0, 0)$$

$$C(0, 1, 0)$$

$$D(0, 0, 1)$$

with respect to the first frame (K_A)

K_B :



$$B(0, 0, 0)$$

$$C(0, 1, 0)$$

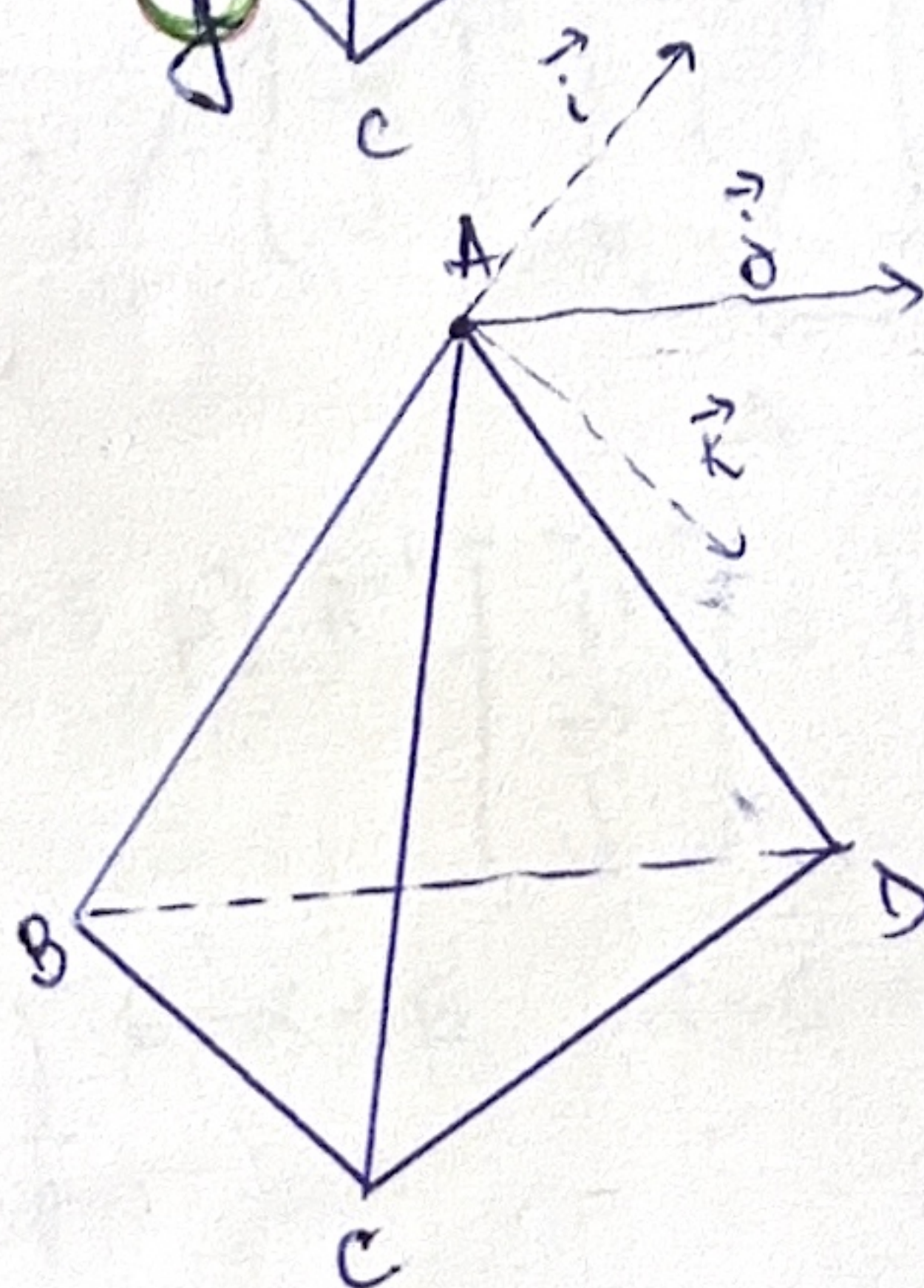
$$D(0, 0, 1)$$

$$A(1, 0, 0)$$

parallel?

$$A(0, 0, 0)$$

K_A' :



$$\vec{AB} = -\vec{i} \Rightarrow B(-1, 0, 0)$$

$$\vec{AC} = \vec{AB} + \vec{BC} = -\vec{i} + \vec{k}$$

$$\Rightarrow C(-1, 0, 1)$$

$$\vec{AD} = \vec{AB} + \vec{BD} = -\vec{i} + \vec{j}$$

$$\Rightarrow D(-1, 1, 0)$$